SOKOINE UNIVERSITY OF AGRICULTURE

SCHOOL OF ENGINEERING AND TECHNOLOGY (SoET)

DEPARTMENT OF AGRICULTURAL ENGINEERING (DAE)

AE 219

ASSIGNMENT 3

Note:

[1] Please listen to videos https://www.youtube.com/playlist?list=PLmKnXyyknMkPhutqZ6uebPazrUCobJ8Ep provided by my Lab YEESI Lab and instructed by Dr Masamaki

[2] Please use Google Colab: https://colab.research.google.com/

[3] Please read and listen to the videos at https://towardsdatascience.com/best-time-saving-python-tricks-in-jupyter-notebook-fca877132507

[4] Please read how to input data into a program: https://www.askpython.com/python/examples/python-user-input

Questions (All programs should run on the command line. Read Instruction 4):

1. **Write a Python program to find the transpose of a matrix (interchange row & column)**
   - ***1.1.****Allow the user to Enter the order of the matrix*
   - ***1.2.****Allow user to Enter the co-efficient of the matrix*
   - ***1.3.****Display both matrix and its transpose*
2. **Write Python program to convert binary to decimal.**
   - ***2.1.****Allow user to Enter the binary number(1s and 0s)*
   - ***2.2.****Display binary and its' decimal equivalent*
3. **Write Python program to check the equality of two matrix**
   - ***3.1.****Allow user to enter the order of the matrix A and B*
   - ***3.2.****Check the two if they are equal*
4. **Write Python Program to Concatenate 2 strings without strcat function**
   - ***4.1.****Allow user to Enter the first and second string then concatenate them without using inbuilt strcat*
   - ***4.2.****Display concatenated string*

5. **Write a Python program to find vowels and consonants in the given sentence**
   - ***5.1.*** *Allow user to Enter a sentence*
   - ***5.2.*** *Calculate and Display No. of vowels and consonants and then display the No. of vowels and consonants.*
6. **Write a Python program to generate and print first N numbers of the following series 1, 2, 5, 9, 16, 27, 45…**
   - ***6.1.*** *Allow user to Enter N so as to specify how long the series should be. E.g. If the user specifies 4 then the program should print only 1, 2, 5, 9. Note that the user should only be allowed to request a series of more than 3 elements.*
7. **Write a Python program to process and display hourly-paid cleaning employees' payroll for weekly payments. Use Python in-built data structure like list to hold your data variables. Assume the program processes payments for one week only**
   - ***7.1.*** The program should accept names, addresses and ages of only 30 employees {Name, address, and age}. The program should only accept employees who are 21 years old or more. **[**
   - ***7.2.*** The program should accept hours worked for each employee every day for seven days from Monday to Sunday. Employees are allowed to work for more than 8 hours per day. Extra time is paid separately.
   - ***7.3.*** The program should calculate the weekly salary for each employee. Assume every hour the worker receives Tsh 4,000 while for an overtime hour is Tsh 6000.
   - ***7.4.*** The program should display the names of cleaning employees and weekly salaries in descending order of the salary.
   - ***7.5.*** The program should be able to show/display the best day in which most employees accumulated the most hours
   - ***7.6.*** Note that >> The program should have a menu to allow the employer to do the above five tasks

Submit your code results by Wednesday midnight (9-June-2022) to the CR as the softcopy and CR will forward the bundle of all codes to kadefue@sua.ac.tz. DO NOT SEND YOUR WORK to Instructor's email. Any enquiries, please send an email.

As usual, Every one hour late equates to 5 marks deduction.